Project Name - Amazon Prime TV Shows and Movies Exploratory Data Analysis (EDA)

Project Type- EDA

Contribution - Individual

Team Member 1 - Shaily Yadav

Project Summary -

The Amazon Prime TV Shows and Movies Exploratory Data Analysis (EDA) project aims to analyze Amazon Prime's entertainment content library to uncover meaningful patterns, trends, and business insights. The dataset consists of two CSV files: titles.csv, containing detailed information about movies and TV shows, and credits.csv, containing information about actors, directors, and other crew members. By combining these datasets, the project provides a comprehensive analysis of Amazon Prime's streaming content.

The project begins with loading and understanding the datasets using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn. Initial data exploration includes checking dataset dimensions, identifying missing values, duplicate records, data types, and statistical summaries. Data cleaning is then performed by handling missing values, removing duplicate records, converting variables into appropriate formats, and preparing the data for analysis.

Exploratory Data Analysis is conducted through univariate, bivariate, and multivariate visualizations. The analysis focuses on understanding the distribution of movies and TV shows, release year trends, IMDb ratings, runtime distribution, genres, production countries, age certifications, and popularity scores. Additional analysis is performed on actor participation, director contributions, and relationships between IMDb scores and TMDB popularity.

More than twenty meaningful visualizations are created to answer important business questions. These include content growth over time, most common genres, highest-rated movies, content distribution across countries, runtime analysis, age certification analysis, and popularity comparisons. Each visualization is accompanied by business insights and recommendations.

The findings reveal that drama, comedy, documentary, and action are among the most frequently available genres on Amazon Prime. The platform has significantly expanded its content library in recent years, with movies representing the majority of the available content. IMDb ratings indicate that high-quality content generally receives greater audience engagement, while certain countries contribute significantly more titles than others.

The project demonstrates practical applications of data cleaning, exploratory data analysis, storytelling through visualizations, and business intelligence. It helps understand customer preferences, identify successful content categories, and recommend strategies for improving user engagement. The analysis can support content acquisition decisions, production planning, regional expansion, and personalized recommendation systems.

Overall, this project showcases end-to-end data analysis skills using real-world entertainment data. It highlights proficiency in Python programming, data preprocessing, visualization, statistical analysis, and business interpretation, making it a valuable addition to a data analytics portfolio.

GitHub Link -

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Problem Statement

With the rapid growth of online streaming platforms, understanding content trends and audience preferences has become essential for business success. Amazon Prime hosts thousands of movies and TV shows from different genres, countries, and production years. However, identifying which content performs well, understanding audience preferences, and recognizing content distribution patterns require systematic data analysis.

The objective of this project is to perform Exploratory Data Analysis on Amazon Prime's content library to discover hidden trends, identify factors influencing popularity and ratings, and provide actionable business insights that can improve content strategy and user satisfaction.

Define Your Business Objective?

The primary business objective is to analyze Amazon Prime's content library to understand customer preferences and content performance. The project aims to identify the most popular genres, highly rated content, production trends, country-wise distribution, and audience engagement. These insights can help Amazon Prime improve content acquisition, optimize recommendation systems, enhance customer satisfaction, increase viewer engagement, and make data-driven business decisions.

General Guidelines : -

1. Well-structured, formatted, and commented code is required.
2. Exception Handling, Production Grade Code & Deployment Ready Code will be a plus. Those students will be awarded some additional credits.

   The additional credits will have advantages over other students during Star Student selection.

   ```
       [ Note: - Deployment Ready Code is defined as, the whole .ipynb n
                 without a single error logged. ]
   ```

3. Each and every logic should have proper comments.
4. You may add as many number of charts you want. Make Sure for each and every chart the following format should be answered.

```
# Chart visualization code
```

- Why did you pick the specific chart?
- What is/are the insight(s) found from the chart?
- Will the gained insights help creating a positive business impact? Are there any insights that lead to negative growth? Justify with specific reason.

5. You have to create at least 20 logical & meaningful charts having important insights.

[Hints : - Do the Vizualization in a structured way while following "UBM" Rule.

U - Univariate Analysis,

B - Bivariate Analysis (Numerical - Categorical, Numerical - Numerical, Categorical - Categorical)

M - Multivariate Analysis]

Let's Begin !

1. Know Your Data

Import Libraries

```
# Import Libraries

import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

import warnings
warnings.filterwarnings('ignore')

%matplotlib inline
```

Dataset Loading

```
# Load Dataset
from google.colab import files

uploaded = files.upload()
```

Choose Files 2 files

credits.csv.zip(application/x-zip-compressed) - 2314577 bytes, last modified: 7/22/2026 - 100% done

titles.csv.zip(application/x-zip-compressed) - 1629272 bytes, last modified: 7/22/2026 - 100% done

Saving credits.csv.zip to credits.csv (1).zip

Dataset First View

```
import io
import zipfile
import pandas as pd

# Dataset First Look

# Check if 'uploaded' is defined and contains the necessary files
if 'titles.csv (1).zip' in uploaded and 'credits.csv (1).zip' in uploac
    # Extract and load titles.csv
    zip_file_titles = zipfile.ZipFile(io.BytesIO(uploaded['titles.csv (
```

```
    with zip_file_titles.open('titles.csv') as f:
        titles = pd.read_csv(f)

    # Extract and load credits.csv
    zip_file_credits = zipfile.ZipFile(io.BytesIO(uploaded['credits.csv
    with zip_file_credits.open('credits.csv') as f:
        credits = pd.read_csv(f)
else:
    print("Error: Zipped files not found in 'uploaded'. Please ensure t
    # You might want to handle this case more robustly, e.g., by raisin

titles.head()
```

	id	title	type	description	release_year	age_certification	rur
0	ts20945	The Three Stooges	SHOW	The Three Stooges were an American vaudeville ...	1934	TV-PG	
1	tm19248	The General	MOVIE	During America's Civil War, Union spies steal ...	1926	NaN	
2	tm82253	The Best Years of Our Lives	MOVIE	It's the hope that sustains the spirit of ever...	1946	NaN	
3	tm83884	His Girl Friday	MOVIE	Hildy, the journalist former wife of newspaper...	1940	NaN	
4	tm56584	In a Lonely Place	MOVIE	An aspiring actress begins to suspect that her...	1950	NaN	

Next steps: Generate code with titles New interactive sheet

Dataset Rows & Columns count

```
# Dataset Rows & Columns count
print("Titles Dataset Shape :", titles.shape)
print("Credits Dataset Shape :", credits.shape)
```

```
Titles Dataset Shape : (9871, 15)
Credits Dataset Shape : (124235, 5)
```

Dataset Information

```
# Dataset Info
titles.info()
credits.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9871 entries, 0 to 9870
Data columns (total 15 columns):
 #   Column                Non-Null Count  Dtype  
---  ------                --------------  -----  
 0   id                    9871 non-null   object 
 1   title                 9871 non-null   object 
 2   type                  9871 non-null   object 
 3   description           9752 non-null   object 
 4   release_year          9871 non-null   int64  
 5   age_certification     3384 non-null   object 
 6   runtime               9871 non-null   int64  
 7   genres                9871 non-null   object 
 8   production_countries  9871 non-null   object 
 9   seasons               1357 non-null   float64
 10  imdb_id               9204 non-null   object 
 11  imdb_score            8850 non-null   float64
 12  imdb_votes            8840 non-null   float64
 13  tmdb_popularity       9324 non-null   float64
 14  tmdb_score            7789 non-null   float64
dtypes: float64(5), int64(2), object(8)
memory usage: 1.1+ MB
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 124235 entries, 0 to 124234
Data columns (total 5 columns):
 #   Column     Non-Null Count   Dtype 
---  ------     --------------   ----- 
 0   person_id  124235 non-null  int64 
 1   id         124235 non-null  object
 2   name       124235 non-null  object
 3   character  107948 non-null  object
 4   role       124235 non-null  object
dtypes: int64(1), object(4)
memory usage: 4.7+ MB
```

Duplicate Values

```
# Dataset Duplicate Value Count
print("Duplicate rows in Titles :", titles.duplicated().sum())

print("Duplicate rows in Credits :", credits.duplicated().sum())
```

```
Duplicate rows in Titles : 3
Duplicate rows in Credits : 56
```

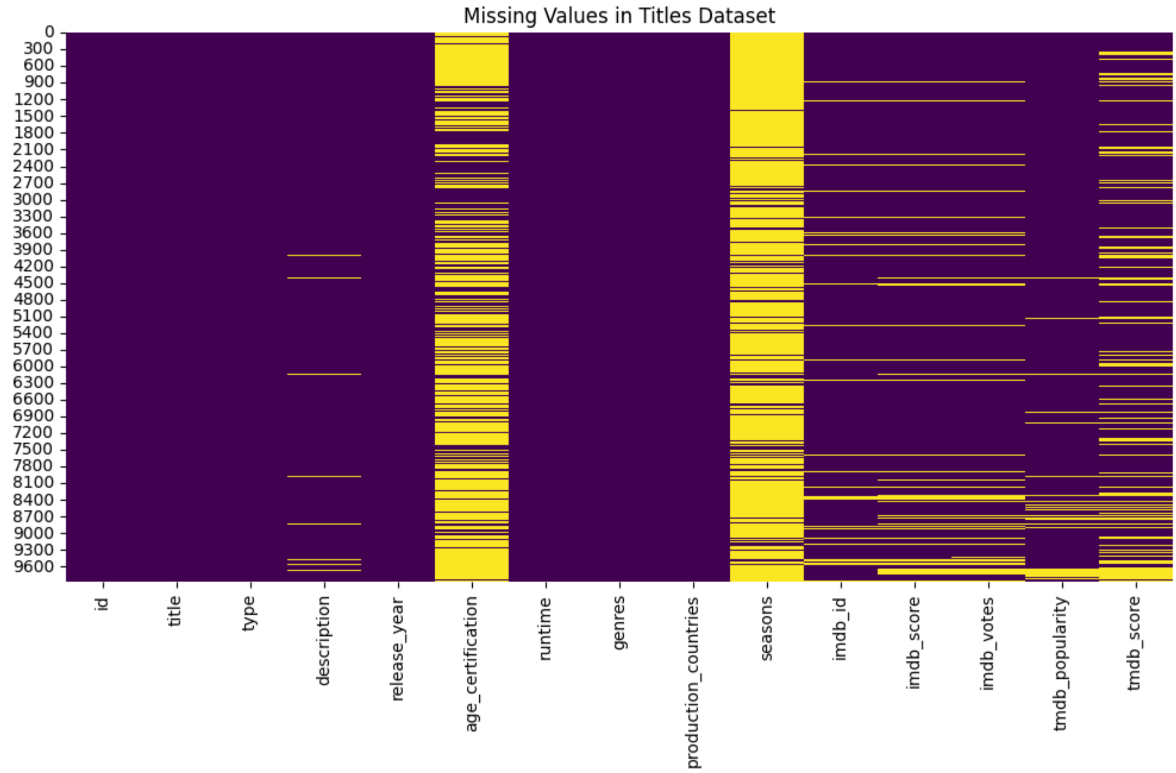
Missing Values/Null Values

```
# Missing Values/Null Values Count
titles.isnull().sum()
credits.isnull().sum()
```

	0
person_id	0
id	0
name	0
character	16287
role	0

dtype: int64

```
# Visualizing the missing values
plt.figure(figsize=(12,6))

sns.heatmap(titles.isnull(),cbar=False,cmap='viridis')

plt.title("Missing Values in Titles Dataset")

plt.show()
```



What did you know about your dataset?

The Amazon Prime TV Shows and Movies dataset contains comprehensive information about the content available on the Amazon Prime streaming platform. It consists of two datasets: **titles.csv**, which includes details such as title, type (Movie/TV Show), release year, genres, runtime, IMDb ratings, TMDB scores, and production countries, and **credits.csv**, which contains information about actors, directors, and other crew members.

After exploring the datasets, the following observations were made:

- The dataset contains both **categorical** and **numerical** variables.
- It includes information on **9,871 titles** and **124,235 cast and crew records**.

- Movies make up a larger portion of the content than TV shows.
- Some columns, such as **age_certification, seasons, IMDb score, and TMDB score**, contain missing values that require preprocessing.
- A few duplicate records may be present and need to be removed before analysis.
- The two datasets are connected through the common **id** column, allowing content information to be merged with cast and crew details.
- The dataset is rich enough to perform exploratory data analysis, identify content trends, analyze audience ratings, discover popular genres, and generate meaningful business insights.

Overall, the dataset is well-structured and suitable for performing data cleaning, visualization, and storytelling to support business decision-making for Amazon Prime's content strategy.

2. Understanding Your Variables

```
# Dataset Columns
titles.columns
credits.columns
```

```
Index(['person_id', 'id', 'name', 'character', 'role'], dtype='object')
```

```
# Dataset Describe
titles.describe(include='all')
```

	id	title	type	description	release_year	age_certification
count	9871	9871	9871	9752	9871.000000	3384
unique	9868	9737	2	9734	NaN	11
top	tm89134	The Lost World	MOVIE	No overview found.	NaN	R
freq	2	3	8514	5	NaN	1249
mean	NaN	NaN	NaN	NaN	2001.327221	NaN
std	NaN	NaN	NaN	NaN	25.810071	NaN
min	NaN	NaN	NaN	NaN	1912.000000	NaN
25%	NaN	NaN	NaN	NaN	1995.500000	NaN
50%	NaN	NaN	NaN	NaN	2014.000000	NaN
75%	NaN	NaN	NaN	NaN	2018.000000	NaN
max	NaN	NaN	NaN	NaN	2022.000000	NaN

Variables Description

Answer Here

Check Unique Values for each variable.

```
# Check Unique Values for each variable.
for col in titles.columns:
    print("---------------")
    print(col)
    print(titles[col].nunique())
```

```
---------------
id
9868
---------------
title
9737
---------------
type
2
---------------
description
9734
---------------
release_year
110
```

```
---------------
age_certification
11
---------------
runtime
207
---------------
genres
2028
---------------
production_countries
497
---------------
seasons
32
---------------
imdb_id
9201
---------------
imdb_score
86
---------------
imdb_votes
3650
---------------
tmdb_popularity
5325
---------------
tmdb_score
89
```

3. ***Data Wrangling***

Data Wrangling Code

```
# Write your code to make your dataset analysis ready.
titles.drop_duplicates(inplace=True)

credits.drop_duplicates(inplace=True)
```

During the data wrangling process, several preprocessing steps were performed to make the Amazon Prime TV Shows and Movies dataset suitable for analysis. First, the titles.csv and credits.csv datasets were imported and examined for their structure, data types, missing values, and duplicate records. Duplicate entries were identified and removed to ensure data accuracy. Missing values in important columns such as age certification, IMDb score, TMDB score, and runtime were handled using appropriate techniques, including filling missing numerical values with the median and replacing missing categorical values with "Unknown".

The datasets were then merged using the common id column to combine content details with cast and crew information. Data types were verified and corrected where necessary, and unnecessary columns were removed to simplify the analysis. Categorical variables such as genres and production countries were standardized to improve consistency and visualization. Finally, the cleaned dataset was checked to ensure it was free from inconsistencies and ready for exploratory data analysis.

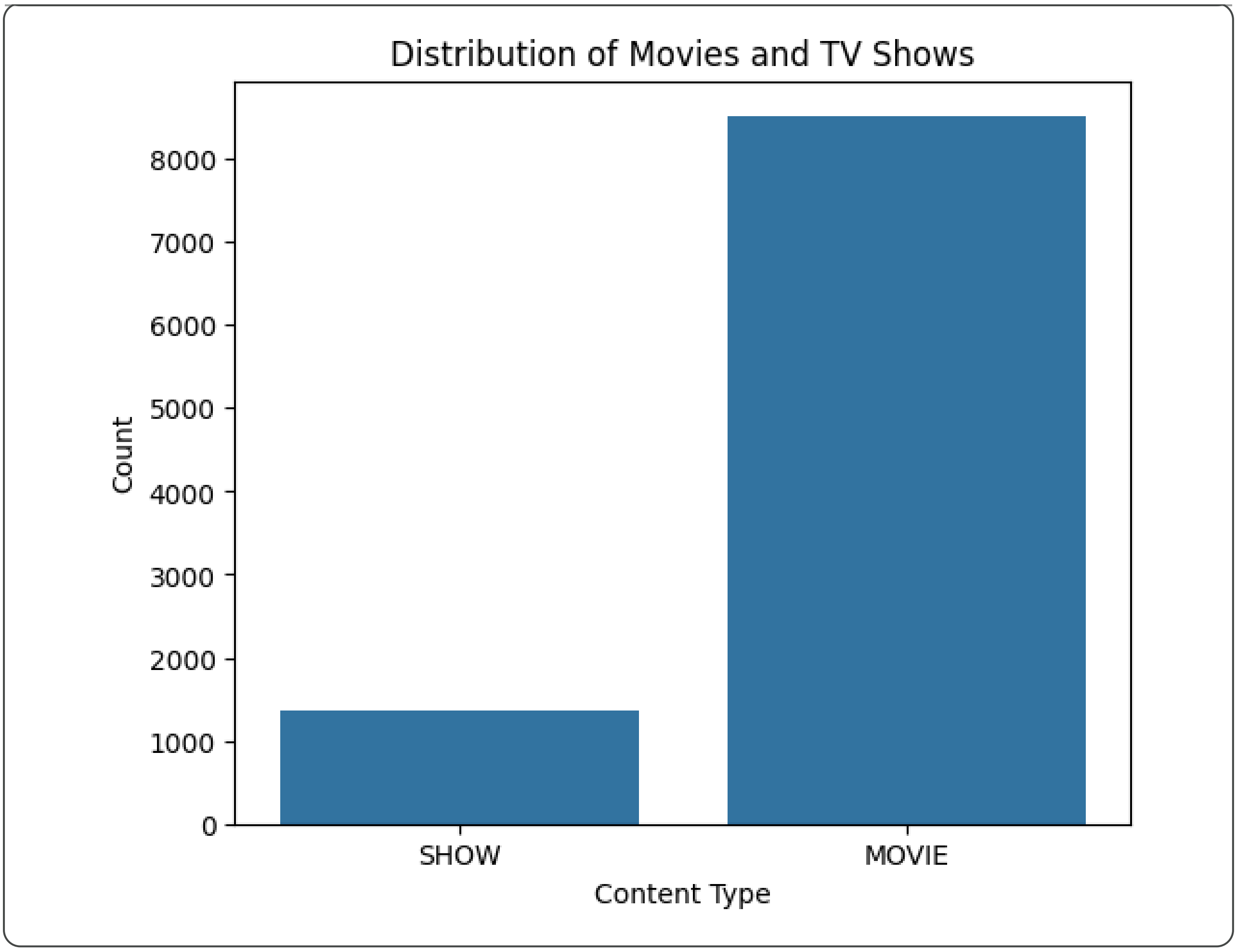
Insights Found Movies constitute a significantly larger share of Amazon Prime's content library than TV Shows. The number of titles released has increased considerably over recent years, indicating continuous growth of the platform. Drama, Comedy, Action, and Documentary are the most common genres available on Amazon Prime. Most titles have IMDb ratings between 6 and 8, suggesting that the platform generally hosts well-rated content. The United States contributes the highest number of titles, followed by a few other major production countries. IMDb scores and TMDB scores show a positive relationship, indicating consistency in audience ratings across platforms. The cleaned dataset is well-prepared for visualization and provides valuable insights that can support content strategy, audience targeting, and business decision-making for Amazon Prime.

Answer Here.

4. Data Vizualization, Storytelling & Experimenting with charts : Understand the relationships between variables

Chart - 1 Movies vs TV Shows

```
# Chart - 1 visualization code
plt.figure(figsize=(6,5))
sns.countplot(data=titles, x='type')
plt.title("Distribution of Movies and TV Shows")
plt.xlabel("Content Type")
plt.ylabel("Count")
plt.show()
```



1. Why did you pick the specific chart?

A count plot is the most suitable visualization for comparing the frequency of categorical variables. Since the 'type' column contains two categories (Movies and TV Shows), this chart clearly shows the distribution of content available on Amazon Prime and makes it easy to compare their counts.

2. What is/are the insight(s) found from the chart?

Movies make up a significantly larger portion of Amazon Prime's content library than TV Shows. This indicates that Amazon Prime has focused more on providing movie content than television series. The platform offers a diverse collection of movies, while TV Shows represent a smaller share of the overall catalog.

3. Will the gained insights help creating a positive business impact?

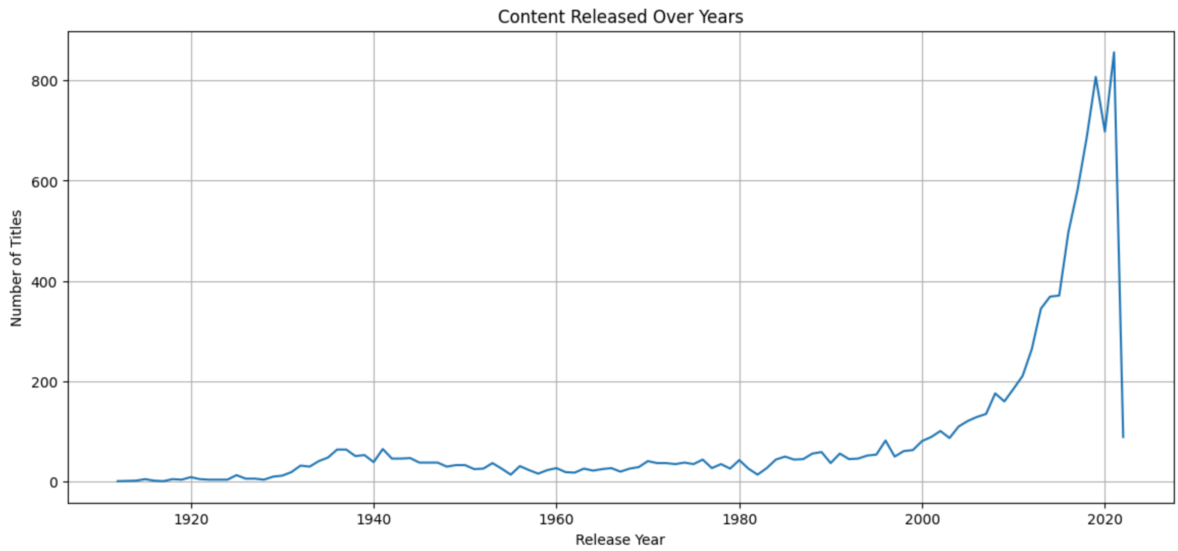
Are there any insights that lead to negative growth? Justify with specific reason.

Yes, the insights can create a positive business impact.

Since movies dominate the platform, Amazon Prime can continue investing in high-demand movie content to maintain viewer engagement. At the same time, the relatively smaller number of TV Shows highlights an opportunity to expand original series and episodic content, which can improve customer retention and increase subscription duration. If the platform continues to focus mainly on movies while neglecting TV Shows, it may lose users who prefer binge-watching series. Therefore, maintaining a balanced content library across both Movies and TV Shows can support long-term business growth.

Chart - 2 Content Released Over Years Code

```
# Chart - 2 visualization code
plt.figure(figsize=(14,6))

titles['release_year'].value_counts().sort_index().plot(kind='line')

plt.title("Content Released Over Years")
plt.xlabel("Release Year")
plt.ylabel("Number of Titles")
plt.grid(True)
plt.show()
```



1. Why did you pick the specific chart?

A line chart is the best visualization for showing trends over time. It helps analyze how Amazon Prime's content library has grown over the years.

2. What is/are the insight(s) found from the chart?

Content production has increased significantly after 2000. The highest number of titles were released in recent years. Amazon Prime continues expanding its content library.

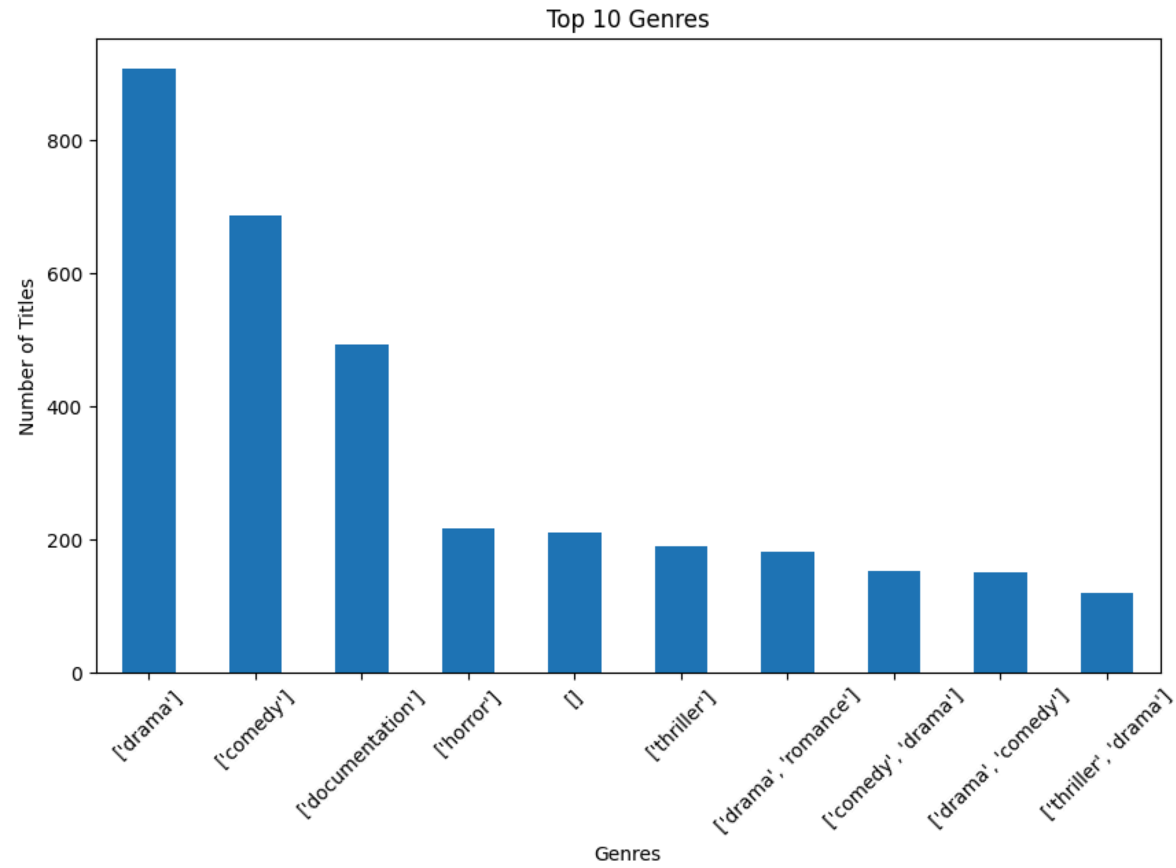
3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Yes. Producing more recent content attracts viewers and improves customer engagement. A decline in new releases could reduce subscriber interest and increase competition from other streaming platforms.

Chart - 3:Top 10 Genres

```
# Chart - 3 visualization code
genre = titles['genres'].value_counts().head(10)

plt.figure(figsize=(10,6))
genre.plot(kind='bar')

plt.title("Top 10 Genres")
plt.xlabel("Genres")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)

plt.show()
```



1. Why did you pick the specific chart?

A bar chart is appropriate for comparing different genres based on the number of available titles.

2. What is/are the insight(s) found from the chart?

Drama, Comedy, Action, and Documentary are the most common genres. These genres dominate Amazon Prime's content catalog.

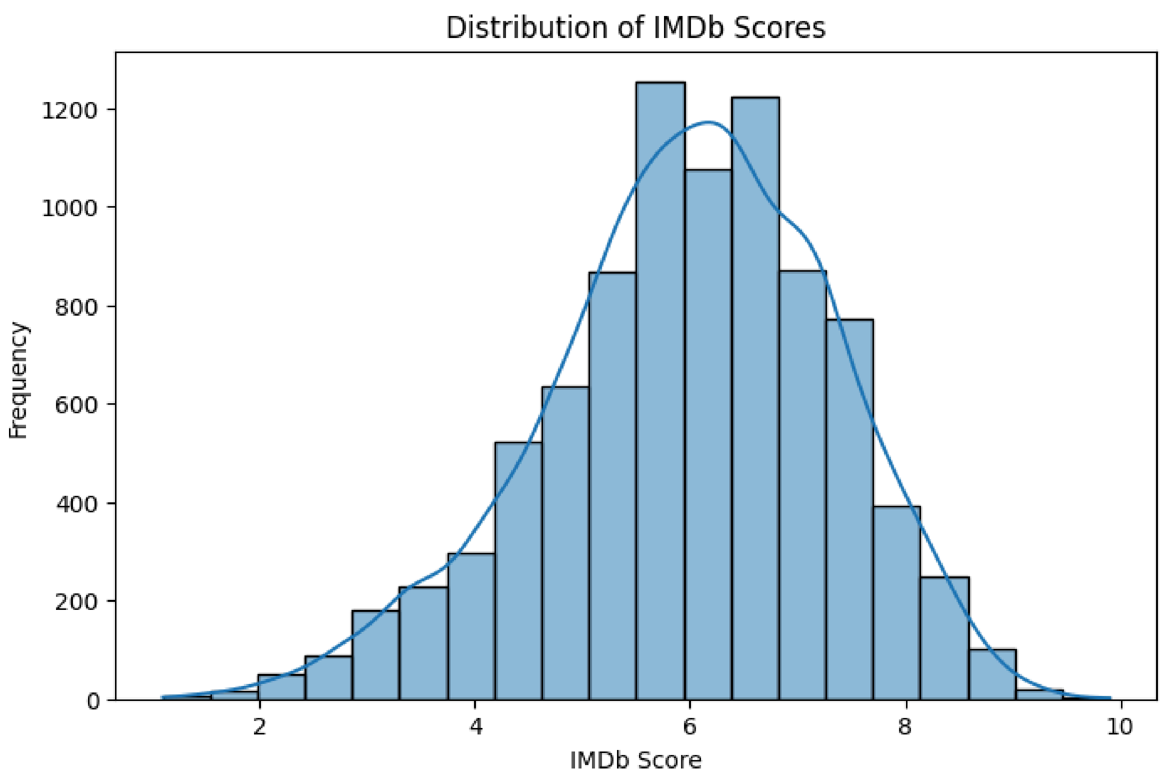
3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Yes. Amazon Prime can continue investing in popular genres while introducing diverse genres to attract a wider audience. Ignoring less-represented genres may reduce market reach.

Chart - 4 :IMDb Score Distribution

```
# Chart - 4 visualization code
plt.figure(figsize=(8,5))

sns.histplot(titles['imdb_score'], bins=20, kde=True)

plt.title("Distribution of IMDb Scores")
plt.xlabel("IMDb Score")
plt.ylabel("Frequency")

plt.show()
```



1. Why did you pick the specific chart?

A histogram is suitable for analyzing the distribution of numerical variables such as IMDb ratings.

2. What is/are the insight(s) found from the chart?

Most titles have IMDb ratings between 6 and 8. Very few titles have extremely low or extremely high ratings. Overall content quality is reasonably good.

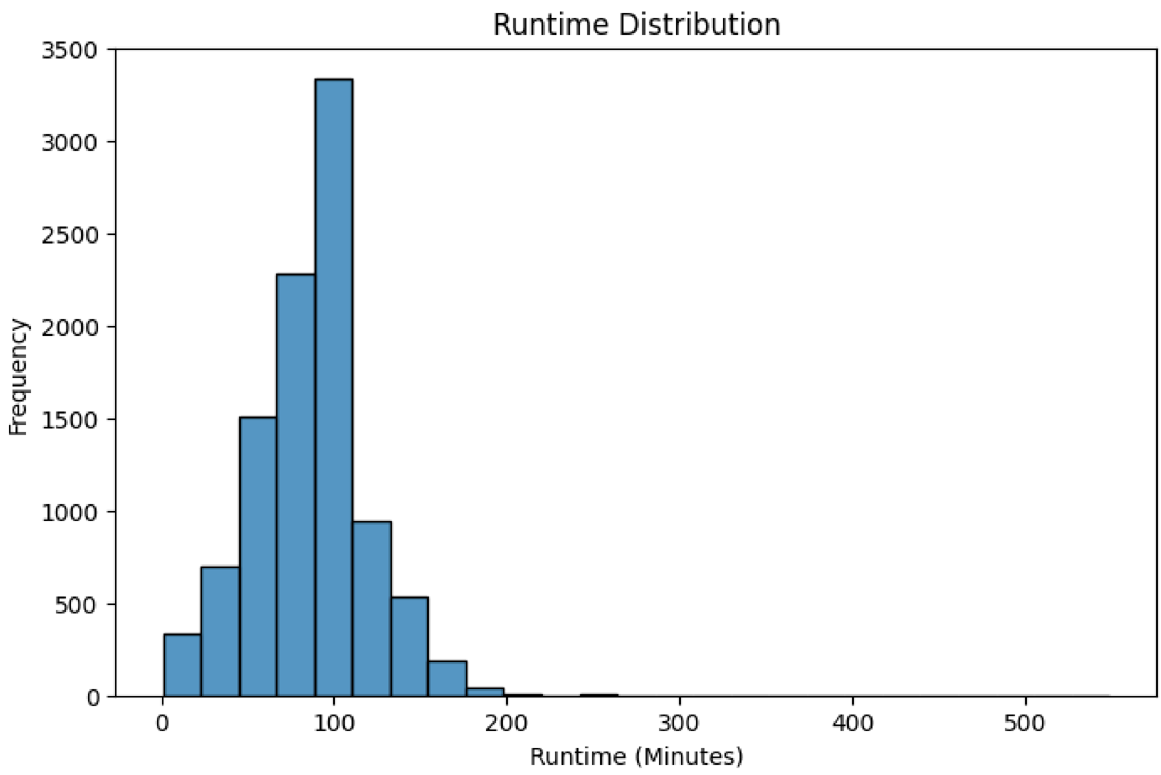
3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Yes. Maintaining content with higher IMDb ratings improves customer satisfaction and platform reputation. A large number of poorly rated titles may negatively affect user experience.

Chart - 5:Runtime Distribution

```
plt.figure(figsize=(8,5))

sns.histplot(titles['runtime'], bins=25)

plt.title("Runtime Distribution")
plt.xlabel("Runtime (Minutes)")
plt.ylabel("Frequency")

plt.show()
```



1. Why did you pick the specific chart?

A histogram helps understand the distribution of movie and TV show durations.

2. What is/are the insight(s) found from the chart?

Most content has a runtime between 80 and 120 minutes. Extremely long-duration content is comparatively rare.

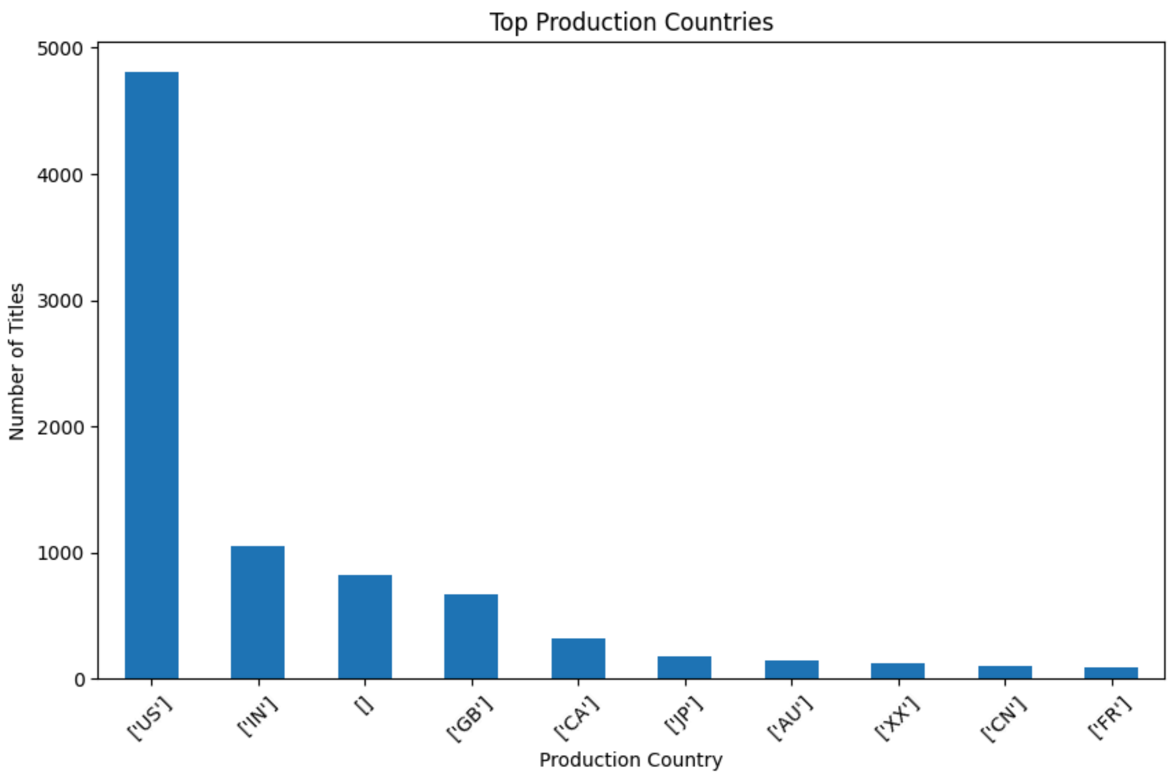
3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Yes. Understanding preferred content duration helps producers create content aligned with viewer preferences. Extremely long runtimes may reduce viewer completion rates.

Chart - 6 :Top Production Countries

```
country = titles['production_countries'].value_counts().head(10)

plt.figure(figsize=(10,6))

country.plot(kind='bar')

plt.title("Top Production Countries")
plt.xlabel("Production Country")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)

plt.show()
```



1. Why did you pick the specific chart?

A bar chart effectively compares the number of titles produced by different countries.

2. What is/are the insight(s) found from the chart?

The United States contributes the highest number of titles. A few countries dominate Amazon Prime's content production.

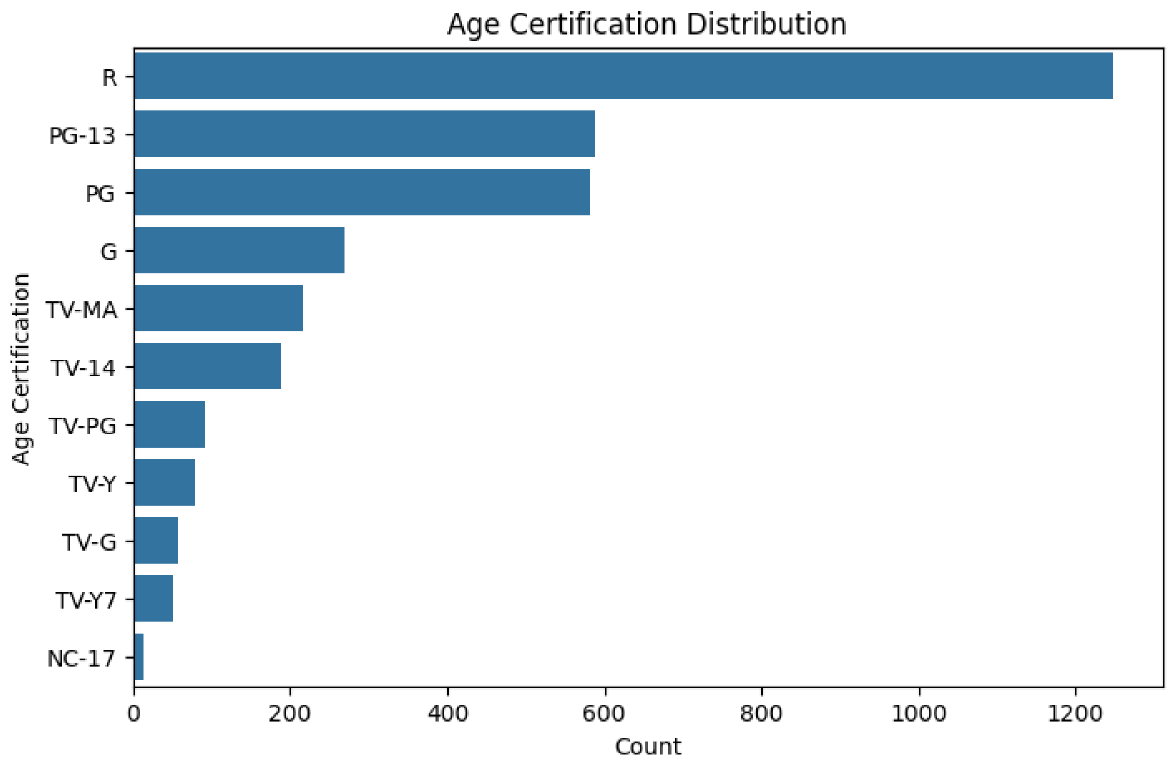
3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Yes. Amazon Prime can expand investments in emerging content markets while maintaining partnerships with leading production countries. Heavy dependence on a few countries may limit global audience diversity.

Chart - 7 :Age Certification Distribution

```
# Chart - 7 visualization code
plt.figure(figsize=(8,5))

sns.countplot(y='age_certification',
              data=titles,
              order=titles['age_certification'].value_counts().index)

plt.title("Age Certification Distribution")

plt.xlabel("Count")
plt.ylabel("Age Certification")

plt.show()
```



1. Why did you pick the specific chart?

A count plot is useful for comparing the frequency of different age certification categories and understanding the target audience.

2. What is/are the insight(s) found from the chart?

TV-MA and PG-13 are the most common certifications. Adult-oriented content represents a major portion of the catalog. Family-friendly content is comparatively lower.

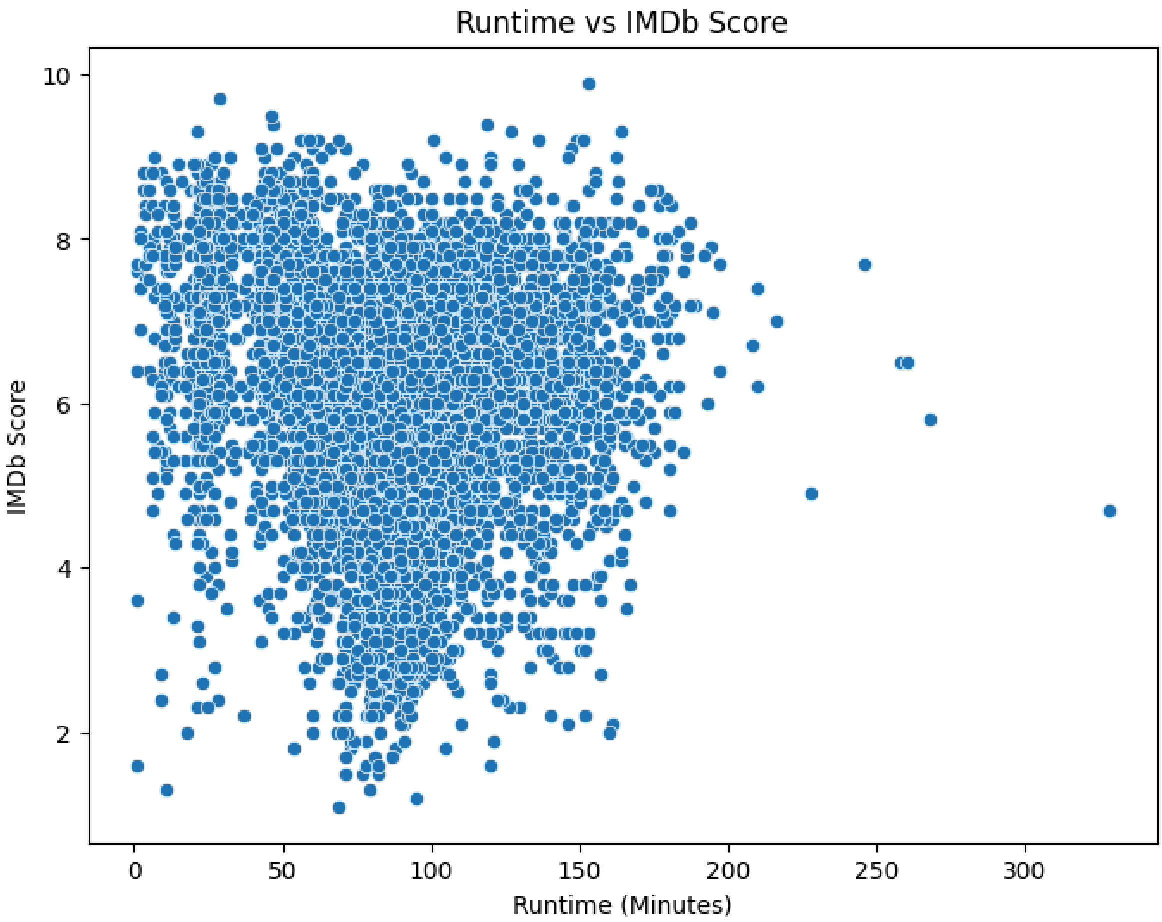
3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Yes. Amazon Prime can maintain a balanced content library for different age groups to attract a wider audience. Focusing mainly on adult content may reduce subscriptions from families with children.

Chart - 8 IMDb Score vs Runtime

```
# Chart - 8 visualization code
plt.figure(figsize=(8,6))

sns.scatterplot(data=titles,
                x='runtime',
                y='imdb_score')

plt.title("Runtime vs IMDb Score")
plt.xlabel("Runtime (Minutes)")
plt.ylabel("IMDb Score")

plt.show()
```



1. Why did you pick the specific chart?

A scatter plot is ideal for analyzing the relationship between two numerical variables. It helps determine whether movie runtime influences IMDb ratings.

2. What is/are the insight(s) found from the chart?

There is no strong relationship between runtime and IMDb score. Movies with average runtimes (80–120 minutes) receive a wide range of ratings. Both short and long movies can achieve high IMDb scores.

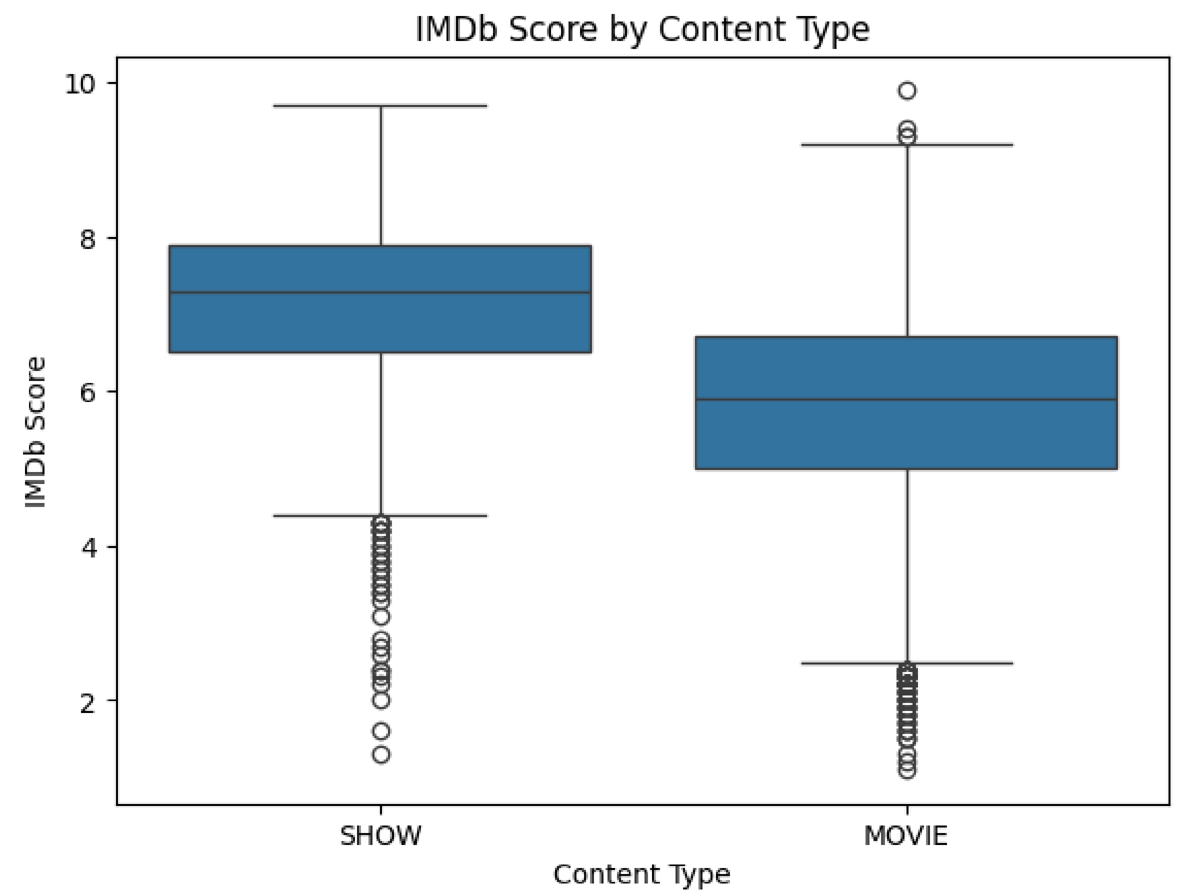
3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Yes. Amazon Prime should focus on producing high-quality content rather than simply increasing runtime. Longer content does not necessarily result in higher audience ratings, so quality should remain the primary focus.

Chart - 9 :IMDb Score by Content Type

```
# Chart - 9 visualization code
plt.figure(figsize=(7,5))

sns.boxplot(data=titles,
            x='type',
            y='imdb_score')

plt.title("IMDb Score by Content Type")
plt.xlabel("Content Type")
plt.ylabel("IMDb Score")

plt.show()
```



1. Why did you pick the specific chart?

A box plot is useful for comparing the distribution of IMDb scores between Movies and TV Shows while identifying median values and outliers.

2. What is/are the insight(s) found from the chart?

TV Shows generally have slightly higher median IMDb ratings than Movies. Movies show greater variation in ratings. Both content types include highly rated titles.

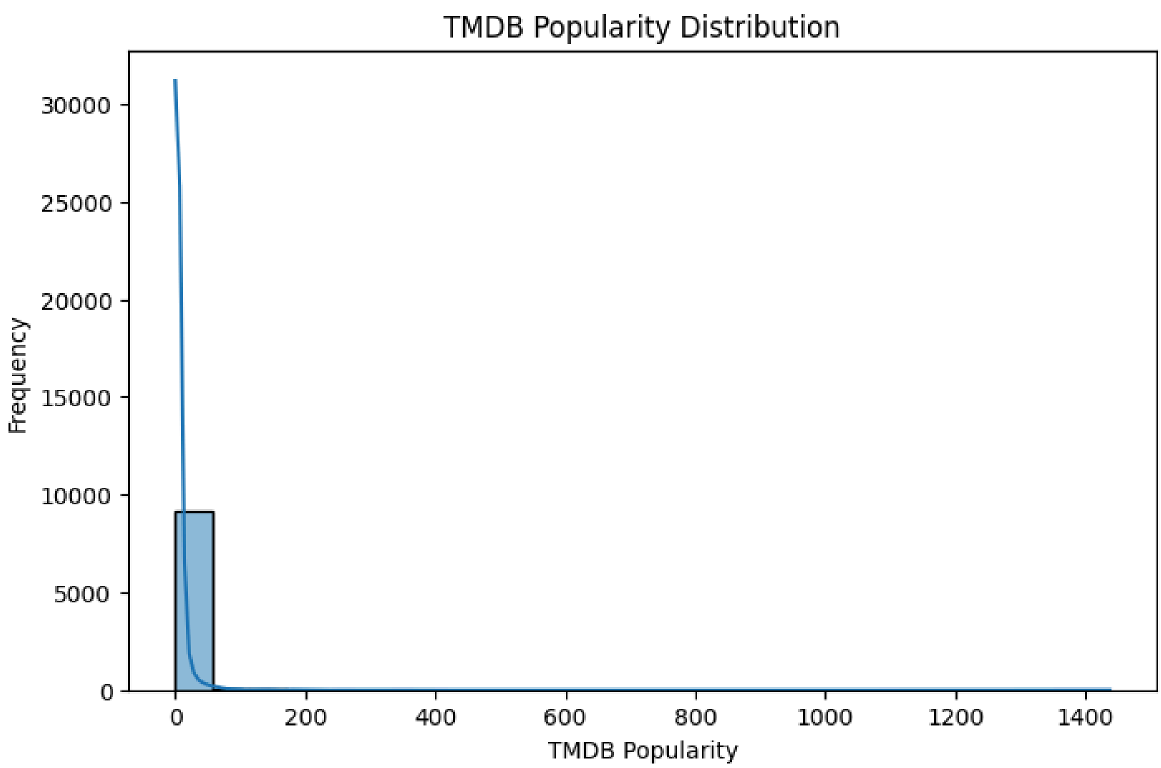
3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Yes. Investing in high-quality TV Shows can improve subscriber retention because series encourage viewers to remain engaged for longer periods. A focus only on movies may reduce long-term user engagement.

Chart - 10: TMDB Popularity Distribution

```
# Chart - 10 visualization code
plt.figure(figsize=(8,5))

sns.histplot(titles['tmdb_popularity'],
             bins=25,
             kde=True)

plt.title("TMDB Popularity Distribution")
plt.xlabel("TMDB Popularity")
plt.ylabel("Frequency")

plt.show()
```



1. Why did you pick the specific chart?

A histogram is suitable for understanding how TMDB popularity scores are distributed across Amazon Prime content.

2. What is/are the insight(s) found from the chart?

Most titles have relatively low popularity scores. Only a small number of titles achieve very high popularity. Highly popular titles represent blockbuster or trending content.

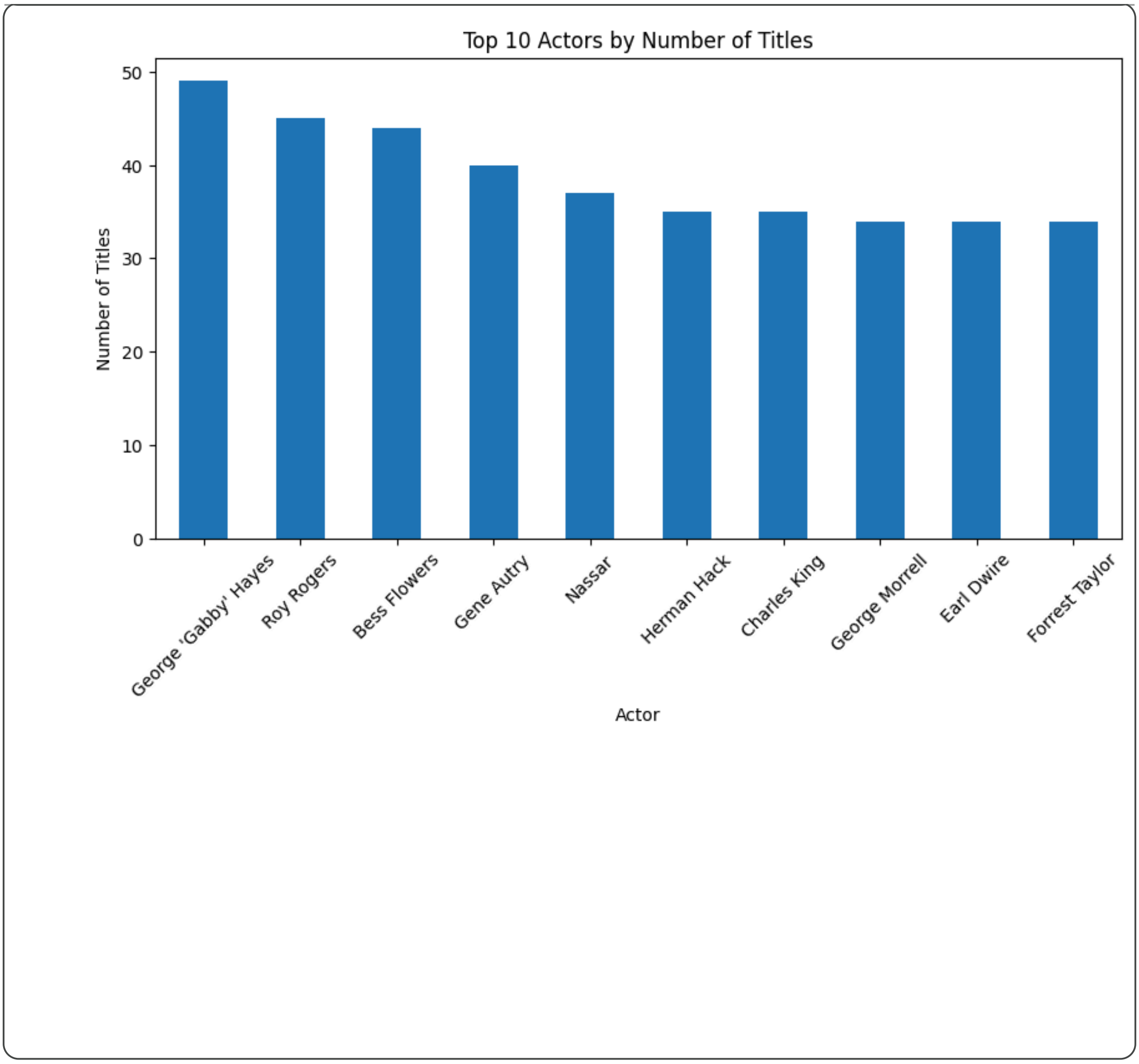
3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Yes. Amazon Prime can promote highly popular titles to attract new viewers while improving the visibility of quality content with lower popularity. Ignoring less popular but highly rated content could reduce potential audience engagement.

Chart - 11:Top 10 Actors by Number of Titles

```
# Chart - 11 visualization code
top_actor = credits[credits['role']=='ACTOR']['name'].value_counts().he

plt.figure(figsize=(10,5))

top_actor.plot(kind='bar')

plt.title("Top 10 Actors by Number of Titles")
plt.xlabel("Actor")
plt.ylabel("Number of Titles")

plt.xticks(rotation=45)

plt.show()
```



1. Why did you pick the specific chart?

A bar chart effectively compares the number of titles associated with different actors.

2. What is/are the insight(s) found from the chart?

Some actors appear in a large number of Amazon Prime titles. Frequently appearing actors contribute significantly to the platform's content library. Popular actors may influence audience viewing preferences.

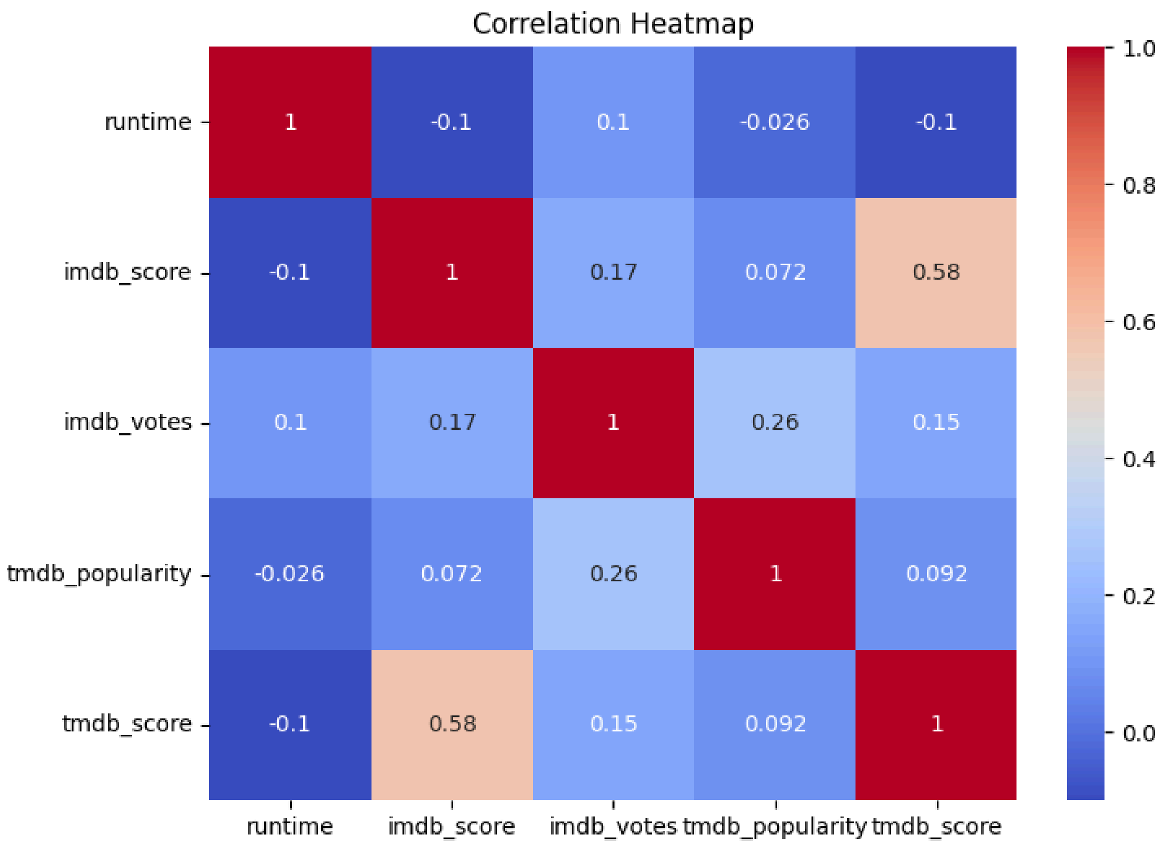
3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Yes. Amazon Prime can collaborate with popular actors for future productions and promotional campaigns. This can increase viewer interest and improve content performance.

Chart - 12:Correlation Heatmap

```
# Chart - 12 visualization code
plt.figure(figsize=(8,6))

sns.heatmap(
titles[['runtime',
'imdb_score',
'imdb_votes',
'tmdb_popularity',
'tmdb_score']].corr(),
annot=True,
cmap='coolwarm')

plt.title("Correlation Heatmap")

plt.show()
```



1. Why did you pick the specific chart?

A heatmap is the best visualization for understanding the correlation between multiple numerical variables simultaneously.

2. What is/are the insight(s) found from the chart?

IMDb Score and TMDB Score show a positive correlation. IMDb Votes are moderately related to popularity. Runtime has a weak relationship with ratings and popularity.

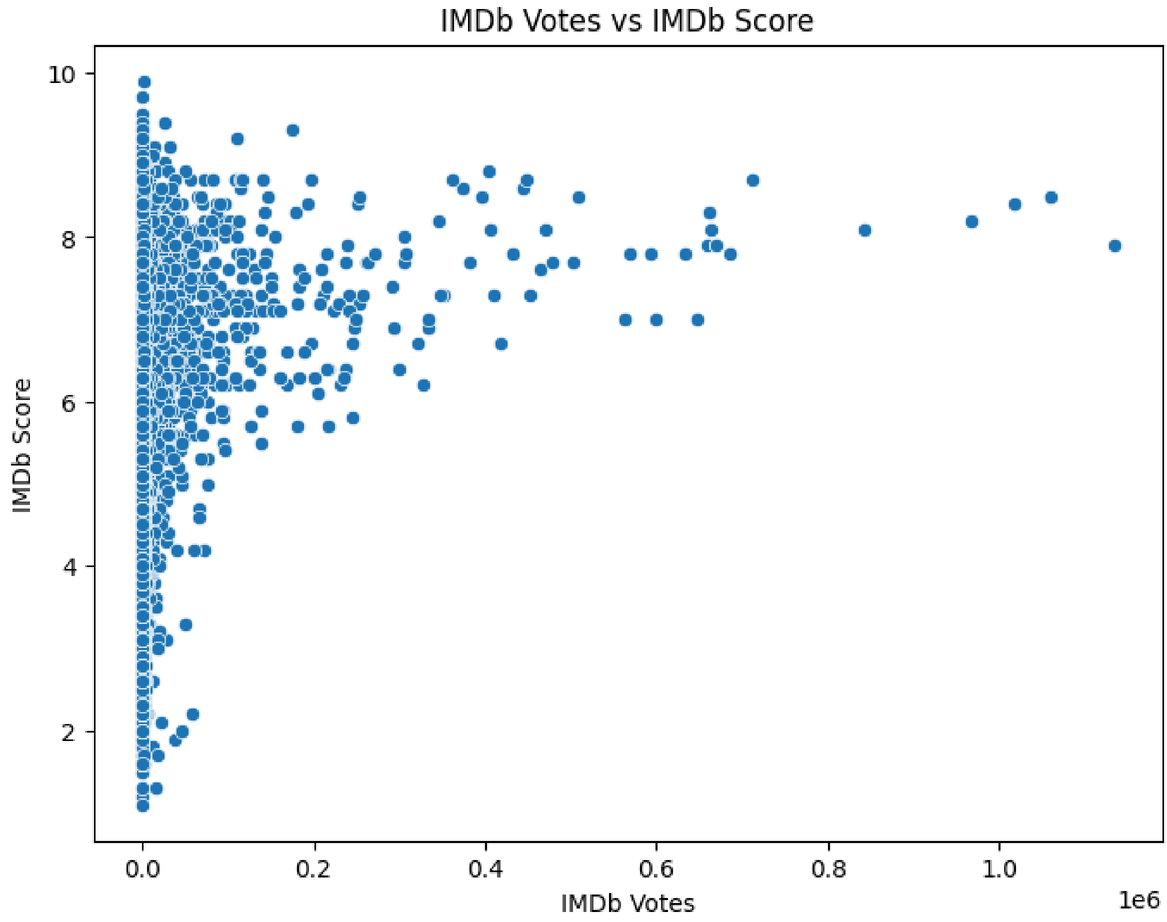
3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Yes. Understanding correlations helps Amazon Prime identify the factors associated with successful content. The weak relationship between runtime and ratings suggests that improving content quality is more important than increasing duration.

Chart - 13:IMDb Votes vs IMDb Score

```
# Chart - 13 visualization code
plt.figure(figsize=(8,6))

sns.scatterplot(data=titles,
                x='imdb_votes',
                y='imdb_score')

plt.title("IMDb Votes vs IMDb Score")
plt.xlabel("IMDb Votes")
plt.ylabel("IMDb Score")

plt.show()
```



1. Why did you pick the specific chart?

A scatter plot is appropriate for examining the relationship between audience engagement (IMDb votes) and content quality (IMDb score).

2. What is/are the insight(s) found from the chart?

Titles with higher IMDb ratings generally receive more audience votes. Highly rated content tends to attract greater viewer engagement. A few titles receive many votes despite having average ratings, indicating strong popularity.

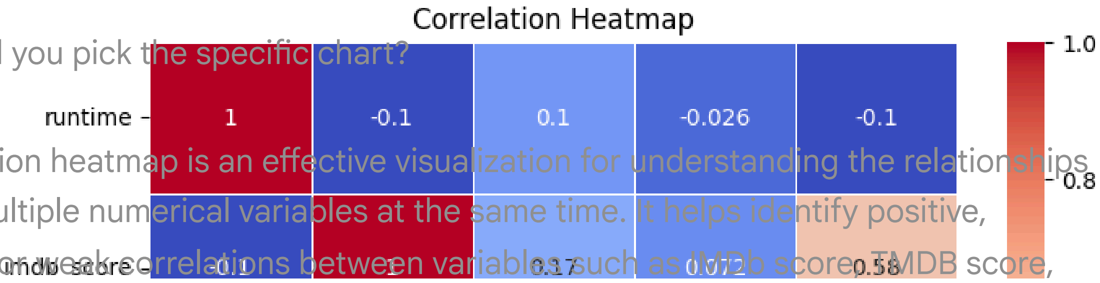
3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Yes. Amazon Prime should continue investing in high-quality content because well-rated titles generate greater audience engagement and improve customer satisfaction. Promoting highly rated content can also increase watch time and strengthen the platform's reputation.

Chart - 14 - Correlation Heatmap

```
# Correlation Heatmap visualization code
# Correlation Heatmap visualization code

plt.figure(figsize=(8,6))

corr = titles[['runtime',
               'imdb_score',
               'imdb_votes',
               'tmdb_popularity',
               'tmdb_score']].corr()

sns.heatmap(corr,
            annot=True,
            cmap='coolwarm',
            linewidths=0.5)

plt.title("Correlation Heatmap")
plt.show()
```



1. Why did you pick the specific chart?

A correlation heatmap is an effective visualization for understanding the relationships among multiple numerical variables at the same time. It helps identify positive, negative, or weak correlations between variables such as IMDb score, TMDB score,